

 Assessment Playback**Assistant**

00:00

Consider a graph search algorithm's OPEN list management. Which of the following statements accurately describe the fundamental differences between Depth-First Search (DFS), Breadth-First Search (BFS), and Best-First Search in terms of their exploration strategy and potential outcomes?

- A. DFS uses a Stack, exploring deeply along one path before backtracking, which may not find the shortest path but is often memory-efficient. BFS uses a Queue, exploring level by level, guaranteeing the shortest path (in unweighted graphs) but can be memory-intensive.
- B. BFS uses a Stack, while DFS uses a Queue. Best-First Search prioritizes nodes based on a heuristic, aiming for a good path but not necessarily the optimal one.
- C. DFS uses a Queue and always finds the optimal solution. BFS uses a Stack and is more memory-efficient than DFS.
- D. Best-First Search uses a Priority Queue to explore nodes with the lowest estimated total cost first, finding optimal solutions and exploring efficiently.

User 00:00

A D

Assistant

00:00

Consider a scenario where you are searching for the shortest path in a graph with uniform edge costs, and the goal state is very close to the start state, but the graph has long (deep) paths as well as has a high branching factor. Which of the following search algorithms would efficiently find the shortest path, and why?

- A. Depth-First Search (DFS) because it explores deeply, quickly reaching potential goal states.
- B. Best-First Search, because it guarantees optimality by expanding minimum cost nodes layer by layer and is well-suited for finding the shortest path when edge costs are uniform.
- C. Breadth-First Search (BFS) that effectively behaving like BFS when edge-weights are uniform.
- D. Depth-First Search (DFS) because it uses less memory than BFS in graphs with high branching factors.

User 00:00

B

Assistant 00:00

Imagine you are trying to solve an ARC-AGI task. You've identified that the problem can be viewed as a search. Which of the following statements accurately reflects how a 'move generator' would function in this context?

- A. The move generator proposes potential transformations (like rotation, translation, or color mapping) that could explain the input-output grid pairs.
- B. The move generator directly modifies individual cells within the output grid until it matches the target.
- C. The move generator enumerates all possible final states for the output grid and checks if any match the training examples.
- D. The move generator focuses on searching for a way to transform the test problem into one of the example outputs..

User 00:00

A B

Assistant 00:00

Consider two sliding tile puzzles with 8 tiles numbered 1 through 8, and one blank space, arranged in a 3x3 grid. If the

tiles are initially in the order 1 2 3 4 5 6 7 8 _ arranged (1) top to bottom, left to right (2) bottom to top, right to left. Which of the following statements is definitively true regarding the solvability of these puzzle to the goal state (blank in the bottom right corner)?

- A. Both puzzles are solvable.
- B. The puzzle (1) is solvable and (2) is unsolvable
- C. The puzzle (2) is solvable and (1) is unsolvable
- D. Both puzzle are unsolvable

User 00:00
A

Assistant 00:00

Consider the 8-tile puzzle. Which of the following invariants can be used to determine if a given initial configuration is solvable, and what is the property of this invariant for solvable states?

- A. The parity of the permutation of the tiles. Solvable states have an even parity.
- B. The sum of the tile numbers. Solvable states have a sum divisible by 3.
- C. The number of tiles out of place. Solvable states have a number of misplaced tiles less than 5.
- D. The parity of the permutation of the tiles. Solvable states have an odd parity.

User 00:00
A

Assistant 00:00

Assessment complete. Thank you.

Recording	Evaluation
Analysis of Graph Search Algorithms (BFS vs. DFS vs. Best-First)	
Grade B	
MULTIPLE CHOICE	
Consider a graph search algorithm's OPEN list management. Which of the following statements accurately describe the fundamental differences between Depth-First Search (DFS), Breadth-First Search (BFS), and Best-First Search (BFS*)? Select all that apply.	

Search (DFS), and Best-First Search in terms of their exploration strategy and potential outcomes?

- A. DFS uses a Stack, exploring deeply along one path before backtracking, which may not find the shortest path but is often memory-efficient. BFS uses a Queue, exploring level by level, guaranteeing the shortest path (in unweighted graphs) but can be memory-intensive.
- B. BFS uses a Stack, while DFS uses a Queue. Best-First Search prioritizes nodes based on a heuristic, aiming for a good path but not necessarily the optimal one.
- C. DFS uses a Queue and always finds the optimal solution. BFS uses a Stack and is more memory-efficient than DFS.
- D. Best-First Search uses a Priority Queue to explore nodes with the lowest estimated total cost first, finding optimal solutions and exploring efficiently.

STUDENT ANSWER

A D

CORRECT ANSWER(S)

A, D

EXPLANATION

Option A correctly identifies DFS's stack-based approach and its trade-offs, and BFS's queue-based approach and its guarantees. Option D accurately describes Best-First Search (like A*) using a priority queue and its benefits. Options B and C contain fundamental inaccuracies about data structures and search properties.

SCORE

5

MULTIPLE CHOICE

Consider a scenario where you are searching for the shortest path in a graph with uniform edge costs, and the goal state is very close to the start state, but the graph has long (deep) paths as well as has a high branching factor. Which of the following search algorithms would efficiently find the shortest path, and why?

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- B. Best-First Search, because it guarantees optimality by expanding minimum cost nodes layer by layer and is well-suited for finding the shortest path when edge costs are uniform.
- C. Breadth-First Search (BFS) that effectively behaving like BFS when edge-weights are uniform.
- D. Depth-First Search (DFS) because it uses less memory than BFS in graphs with high branching factors.

STUDENT ANSWER

B

CORRECT ANSWER(S)

B, C

EXPLANATION

BFS guarantees finding the shortest path in unweighted graphs (uniform edge costs) because it explores all nodes at a given depth before moving to the next depth. Best-First Search with a heuristic of 0 also degenerates to BFS, thus guaranteeing optimality. DFS can get stuck exploring a very deep branch and might not find the shortest path efficiently, and it doesn't guarantee optimality. While DFS can use less memory than BFS in certain deep graphs, its efficiency in finding the shortest path is not guaranteed here.

SCORE

2.5

Modeling ARC-AGI as a Search Problem

Grade D

MULTIPLE CHOICE

Imagine you are trying to solve an ARC-AGI task. You've identified that the problem can be viewed as a search. Which of the following statements accurately reflects how a 'move generator' would function in this context?

- A. The move generator proposes potential transformations (like rotation, translation, or color mapping) that could explain the input-output grid pairs.
- B. The move generator directly modifies individual cells within the output grid until it matches the target.
- C. The move generator enumerates all possible final states for the output grid and checks if any match the training examples.
- D. The move generator focuses on searching for a way to transform the test problem into one of the example outputs..

STUDENT ANSWER

A B

CORRECT ANSWER(S)

A

EXPLANATION

A move generator's role is to infer and propose plausible transformations that could explain the relationship between input and output grids.

SCORE

2.5

Feasibility of 8 Tiles

Grade C

MULTIPLE CHOICE

Consider two sliding tile puzzles with 8 tiles numbered 1 through 8, and one blank space, arranged in a 3x3 grid. If the tiles are initially in the order 1 2 3 4 5 6 7 8 _ arranged (1) top to bottom, left to

right (2) bottom to top, right to left. Which of the following statements is definitively true regarding the solvability of these puzzle to the goal state (blank in the bottom right corner)?

- A. Both puzzles are solvable.
- B. The puzzle (1) is solvable and (2) is unsolvable
- C. The puzzle (2) is solvable and (1) is unsolvable
- D. Both puzzles are unsolvable

STUDENT ANSWER

A

CORRECT ANSWER(S)

D

EXPLANATION

For an $N \times N$ sliding tile puzzle where N is odd, solvability depends on the parity of the number of inversions. For a 3×3 grid ($N=3$, which is odd), the puzzle is solvable if and only if the number of inversions is even. In both cases the total inversions is odd (9 for (1) and 19 for (2)).

SCORE

0

MULTIPLE CHOICE

Consider the 8-tile puzzle. Which of the following invariants can be used to determine if a given initial configuration is solvable, and what is the property of this invariant for solvable states?

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- B. The sum of the tile numbers. Solvable states have a sum divisible by 3.
- C. The number of tiles out of place. Solvable states have a number of misplaced tiles less than 5.
- D. The parity of the permutation of the tiles. Solvable states have an odd parity.

STUDENT ANSWER

A

CORRECT ANSWER(S)

A

EXPLANATION

The parity of the permutation of the tiles, along with the position of the empty tile, is the fundamental invariant used to determine solvability in the 8-tile puzzle. Solvable states must have an even permutation parity.

SCORE

0

57.1%

Needs Improvement

Overall Score

100%

High

Honesty Score

Assessment completed across 3 skills. Skill scores: Analysis of Graph Search Algorithms (BFS vs. DFS vs. Best-First): 73.8/100, Modeling ARC-AGI as a Search Problem: 50.0/100, Feasibility of 8 Tiles: 47.4/100.